

Guideline Title: Targeted Temperature Management

Guideline Number: GUID-3203-C009

Issue date: 09/09/2014	Replaces Dept. Policy: Induced Therapeutic Hypothermia (Post ROSC) GUID-3203-C009
Revision dates: 4/30/24	Developed by: Air Care Team
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Signature: 	Signature:
Department Numbers: 3203	

I. **PURPOSE:** To provide neuroprotection for adult, non-traumatic, non-hemorrhagic, post-cardiac arrest patients once return of spontaneous circulation is achieved. Target temperature management has been demonstrated to improve outcomes in certain patients after resuscitation from cardiac arrest, but early initiation of this treatment has not proven beneficial. Consult with accepting physician prior to administering this treatment.

II. **DEFINITIONS:**

- A. VF - Ventricular Fibrillation
- B. VT - Ventricular Tachycardia
- C. PEA - Pulseless Electrical Activity
- D. CPR – Cardio-Pulmonary Resuscitation
- E. ACLS – Advanced Cardiac Life Support
- F. GCS – Glasgow Coma Scale

III. **DEPARTMENT GUIDELINE:**

- A. Continue strict temperature management initiated by sending physician (The temperature goal is 33°C).
- B. Inclusion Criteria (REQUIRES ALL FOUR COMPONENTS)
 - 1. Medical Cardiac Arrest: VF, pulseless VT, PEA, & Asystole.
 - 2. Downtime (5-20 minutes): Downtime is defined as: the time from the onset of cardiac arrest to the initiation of ACLS. Bystander CPR is not included.
 - 3. Comatose Patients (GCS < 9): Patients who do not respond appropriately to verbal commands after return of spontaneous circulation. Agitated and combative patients are comatose by this definition and should be cooled.
 - 4. Patients greater than the age 18 are eligible.
- C. Exclusion Criteria (any of the following)
 - 1. CPR for more than 45 minutes.
 - 2. Un-witnessed asystole or PEA, refractory cardiogenic or septic shock despite intravenous fluids and vasopressors.
 - 3. Severe coagulopathy (with clinical evidence of bleeding).

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4. Other causes of coma.
5. Known or suspected pregnancy (consider target temp of 36 degrees).
6. Do Not Resuscitate (DNR) status.
7. Other acute brain injury including traumatic brain injury, acute stroke, overdose, or status epilepticus.
8. Acute major trauma with significant bleeding due to concerns of coagulopathy.
9. Initial assessment temperature < 32°C.
- D. Ensure patient is intubated and on mechanical ventilation if possible. SaO₂ goal 94-96%. Keep HOB > 30°.
- E. Obtain and document a complete neurological exam to include GCS, pupillary size, and response to light, and oculocephalic reflex (dolls eyes).
- F. Consider analgesia & sedation management per *GUID3203-G005 Analgesia and Sedation Management* or *GUID3203-G008 Propofol Infusion* to maintain Richmond Agitation and Sedation Scale (RASS) scale -4.
- G. Monitor temperature by one of the following measures: rectal, foley catheter, or esophageal monitoring device. Document temperature every 15 minutes.
- H. Obtain or verify recent blood glucose level.
- I. If patient begins to shiver, first ensure adequate sedation.
 1. Consider **Rocuronium 50 mg IV/IO** for paralysis.
 2. Wrap head, hands, and feet with towels or blankets.
- J. If hypotension persists, consider using vasopressors per *GUID3203-M011 Hypotension*.
- IV. **DOCUMENTATION:**
 - A. EMS charts
- V. **REFERENCES:**
 - A. American Heart Association, 2015.